

HYPERBOLIC GEOMETRY — EXERCISES FROM LECTURE 2

Exercise 2.A

Consider the sphere

$$\mathbb{S}^2 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{R}^3.$$

Write the Riemannian metric on $\mathbb{S}^2$ induced by the standard Riemannian structure of $\mathbb{R}^3$ in the local coordinates given by stereographic projection.

Exercise 2.B

Prove that the natural projection $\mathbb{R}^d \rightarrow \mathbb{T}^d$ is a Riemannian isometric embedding for the flat structure on $\mathbb{T}^d$. Is it an isometric embedding of metric spaces?

Exercise 2.C

Let $f : (M, g) \rightarrow (N, h)$ be a Riemannian isometric embedding. Prove that, if $\gamma : I \rightarrow M$ is a geodesic in (M, g) , then $f \circ \gamma$ is a geodesic in (N, h) . Deduce that f is a local isometric embedding of metric spaces.

Exercise 2.D

If $\gamma : [0, 1] \rightarrow M$ is a geodesic with $\gamma(0) = \gamma(1) = p$, prove that

$$\text{inj}_p(M) \leq \frac{1}{2}|\gamma|_g.$$

Exercise 2.E

Let (M, g) be a Riemannian manifold.

- (i) Let $\gamma : [a, b] \rightarrow M$ be a C^1 path with constant speed (i.e. there is a constant K such that $\|\gamma'(t)\|_g^{\gamma(t)} = K$ for all $t \in [a, b]$). Suppose that for every C^1 path $\eta : [a, b] \rightarrow M$ with $\gamma(a) = \eta(a)$ and $\gamma(b) = \eta(b)$, we have $|\gamma|_g \leq |\eta|_g$. Prove that γ is a geodesic.
- (ii) Prove that the converse does not hold.
- (iii) Prove the following partial converse: let $p \in M$, and consider $B := \{v \in T_p M : \|v\|_g^p < \text{inj}_p(M)\}$. If $\gamma : [0, 1] \rightarrow \exp_p(B)$ is a geodesic with $\gamma(0) = p$, and $c : [0, 1] \rightarrow M$ is a piecewise C^1 path with $c(0) = \gamma(0)$ and $c(1) = \gamma(1)$, then

$$|c|_g \geq |\gamma|_g.$$